OTH REGENSBURG

Praktikum Regelungstechnik (RTP)

Versuch T3:
RetBot system identification, part I

2026-05-02

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1 Preparations

1.1. (aV) Explanation of pulse width modulation control for a drive

Pulse width modulation (PWM) is a technique used to control the power delivered to a load, such as a motor, by varying the width of the pulses in a pulse train. The duty cycle of the PWM signal determines the average voltage applied to the load, allowing for precise control of the motor's speed and torque. The important parameters of PWM include, the impulse width (the duration of the 'on' time), the period (the total duration of one cycle), the break time (the duration of the 'off' time), and the VCC span (the voltage level of the 'on' and 'off' states) as shown in Figure 1.1.
The duty cycles of the PWM signal can be adjusted to control the motor's behavior. See Figure 1.2 for example

- 10 % duty cycle means the motor receives power for 10 % of the time, and is off for the remaining 90 %.
- 50 % duty cycle means the motor receives power for 50 % of the time, and is off for the remaining 50 %.
- 90 % duty cycle means the motor receives power for 90 % of the time, and is off for the remaining 10 %.

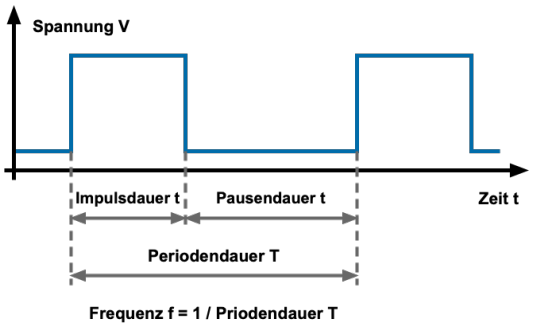


Abbildung 1.1.: Illustration of PWM parameters[1]

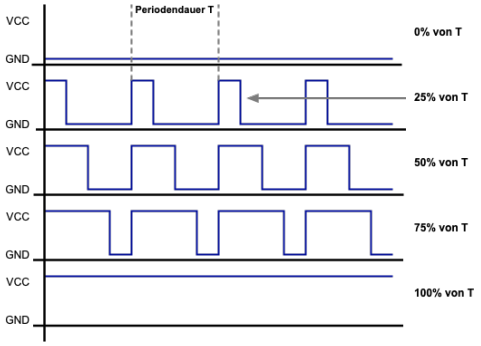


Abbildung 1.2.: Illustration of PWM duty cycles[1]

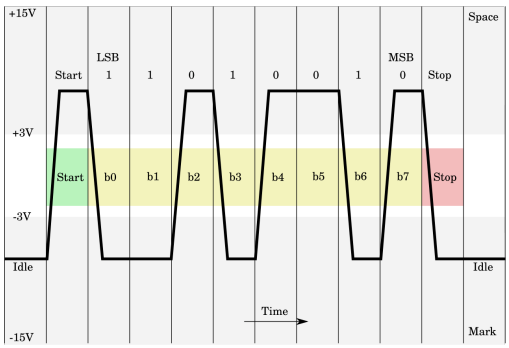


Abbildung 1.3.: By `Rs232_oscilloscope_trace.jpg`: Ktnbn, derivative work Samuel Tardieu (talk), `Rs232_oscilloscope_trace.jpg`, CC SA 1.0, https://commons.wikimedia.org/w/index.php?curid=6101703 [2]

1.2. (bV) Serial interface (RS232) and its key parameters

The RS232 interface is used for connecting devices together one sender and one receiver. Before USB it was one of the most common interfaces for connecting peripherals to computers. Both the sender and the receiver have to be configured with the same parameters for the communication to work. The key parameters are shown in figure 1.3 and include:

- Baud rate: The speed of data transmission, measured in bits per second (bps). Common baud rates include and range from 110, 9600, 19200, 38400, 57600, and 115200 bps.
- Data bits: The number of bits used to represent each character. Common settings are 7 or 8 data bits.
- Parity: A method of error checking. It can be set to none, even, odd, mark, or space.
- Stop bits: The number and configuration of stop bits used to signal the end of a data packet.
- Flow control: A method to control the flow of data between devices, which can be hardware (RTS/CTS) or software (XON/XOFF).
- Voltage levels: RS232 uses specific voltage levels for signaling, typically +3 to +15 volts for a logical '1' and -3 to -15 volts for a logical '0'.

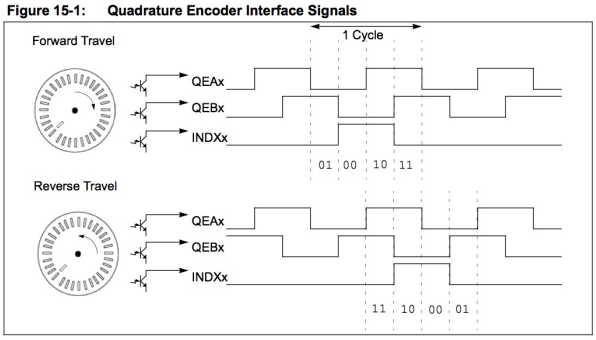


Abbildung 1.4.: Illustration of Quadrature Encoder Interface (QEI) signals[3]

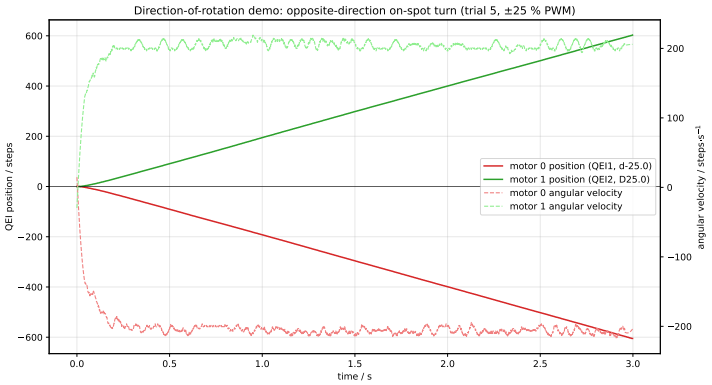
1.3. (cV) Using QEI to detect the direction of rotation

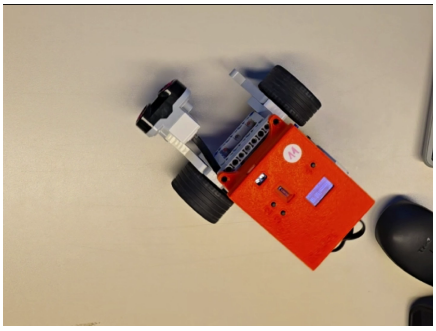
Quadrature Encoder Interface (QEI) is a method used to determine the position and direction of rotation of a motor shaft. It uses two signals, typically referred to as A and B, which are generated by the encoder as the shaft rotates. [3] QEI provides an interface to incremental encoders for obtaining mechanical position and speed data from rotating systems, enabling closed-loop motor control. A typical quadrature encoder consists of a slotted wheel mounted on the motor shaft and an emitter/detector module that senses the slots, producing two phase-shifted output signals, see figure 1.4, Phase A (QEAx) and Phase B (QEBx), along with an Index pulse (INDXx) that marks a reference position once per mechanical revolution. The relative phase between QEAx and QEBx encodes the direction of rotation, while a 16-bit Up/Down Counter (POSxCNT) tracks the angular position by incrementing or decrementing on each state transition. [3]

To illustrate the operating principle, consider a slotted disk mounted on the motor shaft with $N = 360$ evenly spaced slots around its circumference, sensed by two laser-based optical detectors offset by a quarter slot pitch (i.e. 90° electrical phase shift). As the disk rotates, each detector produces a square wave: detector A generates QEAx and detector B generates QEBx, with both signals running at the same frequency but offset in phase. For forward rotation, QEAx leads QEBx by 90°, so the (QEAx, QEBx) pair cycles through the state sequence $00 \rightarrow 10 \rightarrow 11 \rightarrow 01 \rightarrow 00$; for reverse rotation, the sequence is traversed in the opposite order. Each transition between states is counted, yielding $4N = 1440$ counts per mechanical revolution and an angular resolution of 0,25°. At a shaft speed of, for example, 600 rpm (10 Hz), the QEAx and QEBx signals oscillate at 3,6 kHz and the counter POSxCNT updates at 14,4 kHz, incrementing for forward rotation and decrementing for reverse. Once per revolution, the INDXx pulse asserts and can be used to reset POSxCNT, providing an absolute reference that eliminates accumulated drift.

To visualize and confirm the direction of rotation, the experiment described in Chapter 2 has been performed, where the QEI signals were recorded while the motor was

driven in both forward and reverse directions. The resulting QEI identifies the direction of rotation.





https://youtu.be/qWWmoOZy7aw

Abbildung 1.5.: Opposite-direction on-spot turn at ±25 % PWM. Left: motor 0 (red, `d-25.0`) and motor 1 (green, `D25.0`); solid lines show QEI position (left axis), dashed lines show angular velocity (right axis). Right: inline video of the on-spot turn.

1.4. (dV) Review of documentation for Serial.ICE

From the documentation of Serial.ICE the following key information can be extracted:

- Two options to send data: (I) through the direct-input main field, and (II) through the 'Data to send' field.
- Data is received in the 'Main window' field and shown there.
- Logging: in the 'Serial window', click on 'LogtoFile' to log the communication to a file. The file is saved in the same directory as `Serial.ICE.exe`.

The necessary commands are `D25.0`, `D50.0`, `D75.0`, and `D100.0` to set the duty cycle of the PWM signal to 25 %, 50 %, 75 %, and 100 % respectively.
Based on this objective, a standardized test plan is implemented: (i) Define communication parameters (3.0 Mbaud, 8N1, no parity/flow control), (ii) Set the drive to PWM constant mode, (iii) reset the QEI counter and document environmental conditions, (iv) for each speed level, perform a jump from a standstill and fully log the response. This provides reproducible data sets that can subsequently be evaluated in a comparable manner.

2 Experiment T3: RetBot system identification, part I

The EV3 robot was controlled via Serial.ICE using USB/virtual COM port. In constant PWM mode, symmetrical duty cycles of 25 %, 50 %, 75 %, and 100 % were initially set for both drives (`d/D` refer to motors 0/1). To ensure complete traceability, the following standardized procedure was followed:

1. Connect and configure: plug in the EV3 via USB, open Serial.ICE, set communication parameters (e.g. 3,0 Mbaud, 8N1, no parity/flow control), optionally send `v` (version) and `r` (QEI reset).
2. Set PWM and start logging: select mode (`m1`), command desired PWM (`d/D`, e.g. `d50.0`, `D50.0`). Activate the log function in Serial.ICE so that commands and responses are recorded.
3. Positioning and test run: disconnect the USB cable, position the robot as defined on the track. Start the test locally on the device by pressing button 2; press button 2 again to stop. (The controller block centrally controls the inputs and outputs.)
4. Retrieve and save measurement data: reconnect the robot, send the command `g` in Serial.ICE to read the measurement data from the controller. Save/close the log file using the Log button.
5. Repeat for all duty cycles: repeat steps 1 bis 4 for 25 %, 50 %, 75 %, 100 %, and, if applicable, for asymmetric settings (e.g. `d25/D24`, `d50/D47`, `d75/D71`, `d100/D95`) to improve tracking and compensate for motor differences.

2.1. (e) Documentation of communication with the RetBot

The RetBot controller exposes a small single-letter command set over the RS232 link configured in Section 1.3; each command is either acknowledged silently or by a data block written to the Serial.ICE log:

- `r` — reset both QEI counters to zero (no reply).

- `d`*xx.x* and `D`*xx.x* — set the PWM duty cycle of motor 0 and motor 1 respectively to *xx.x* % (no reply; e.g. `d25.0`, `D25.0`).
- `g` — request the buffered measurement record. The controller streams 2250 samples of `Time/ms;QEI1/steps;QEI2/steps;d/cm;Mot0/%` sampled at $f_s = 500\,\text{Hz}$, which Serial.ICE captures to file.

Listing 2.1 reproduces the first $\sim 100\,\text{ms}$ of a `d25.0` run: the `Mot0/%` field switches from 0 to 25 at $t = 40\,\text{ms}$ when the command is issued, and the first non-zero `QEI1` step is recorded $\Delta t = 18\,\text{ms}$ later — the static-friction breakaway dead-time of motor 0.

Listing 2.1: Excerpt of a `g`-command response logged by Serial.ICE (first 100 ms of a `d25.0` run).

```
fsample=500Hz
Time/ms;QEI1/steps;QEI2/steps;d/cm;Mot0/%
   0;   0;   0; 32.0;  0
   2;   0;   0; 32.0;  0
   4;   0;   0; 32.0;  0
   6;   0;   0; 32.0;  0
   8;   0;   0; 32.0;  0
  10;   0;   0; 32.0;  0
  12;   0;   0; 32.0;  0
  14;   0;   0; 32.0;  0
  16;   0;   0; 32.0;  0
  18;   0;   0; 32.0;  0
  20;   0;   0; 32.0;  0
  22;   0;   0; 32.0;  0
  24;   0;   0; 32.0;  0
  26;   0;   1; 32.0;  0
  28;   0;   1; 32.0;  0
  30;   0;   1; 32.0;  0
  32;   0;   1; 32.0;  0
  34;   0;   2; 32.0;  0
  36;   0;   2; 32.0;  0
  38;   0;   2; 32.0;  0
  40;   0;   2; 32.0; 25
  42;   0;   3; 32.0; 25
  44;   0;   3; 32.0; 25
  46;   0;   3; 32.0; 25
  48;   0;   4; 32.0; 25
  50;   0;   4; 32.0; 25
  52;   0;   4; 32.0; 25
  54;   0;   4; 32.0; 25
  56;   0;   5; 32.0; 25
  58;   1;   5; 32.0; 25
  60;   1;   6; 32.0; 25
  62;   1;   6; 32.0; 25
  64;   1;   6; 32.0; 25
  66;   1;   6; 32.0; 25
  68;   2;   7; 32.0; 25
  70;   2;   7; 32.0; 25
  72;   2;   8; 32.0; 25
  74;   2;   8; 32.0; 25
  76;   2;   8; 32.0; 25
  78;   3;   9; 32.0; 25
  80;   3;   9; 32.0; 25
  82;   3;   9; 32.0; 25
  84;   4;  10; 32.0; 25
  86;   4;  10; 32.0; 25
  88;   4;  10; 32.0; 25
  90;   4;  11; 32.0; 25
  92;   5;  11; 32.0; 25
```

```
 94;   5;  12; 32.0; 25
 96;   5;  12; 32.0; 25
 98;   6;  12; 32.0; 25
100;   6;  13; 32.0; 25
```

2.1.1. Observed behavior

All four drives exhibited a consistent leftward drift: motor 1 (right wheel, `QEI2`) accumulated 2–3 % more counts than motor 0 (left wheel, `QEI1`) for the same commanded duty cycle, the gap growing approximately linearly with travel (Figures 2.3 and 2.1). This asymmetry is plausibly attributable to ordinary real-world tolerances — gearbox play, contact-patch and tire-wear differences, bearing friction, supply-voltage drop along the cabling — and motivates the asymmetric trim values listed in the procedure (e.g. `d100/D95`). For subsequent system identification it implies that the two drive trains should be modeled separately rather than as a single symmetric plant.
A more localized feature is visible only in the 25 % response (Figure 2.1): around $t \approx 0{,}75\,\mathrm{s}$ the motor 0 QEI position jumps by several counts in a single sample, so the numerical derivative exhibits a near-vertical spike. The Savitzky-Golay window (Section C.2) spreads this near-discontinuity into a $\sim 40\,\mathrm{ms}$ plateau in the velocity trace. Because no corresponding event appears on motor 1 and the steady-state slope on either side of the plateau is unchanged, the most likely cause is a transient QEI counting glitch or a brief serial-link drop-out rather than a real mechanical event.

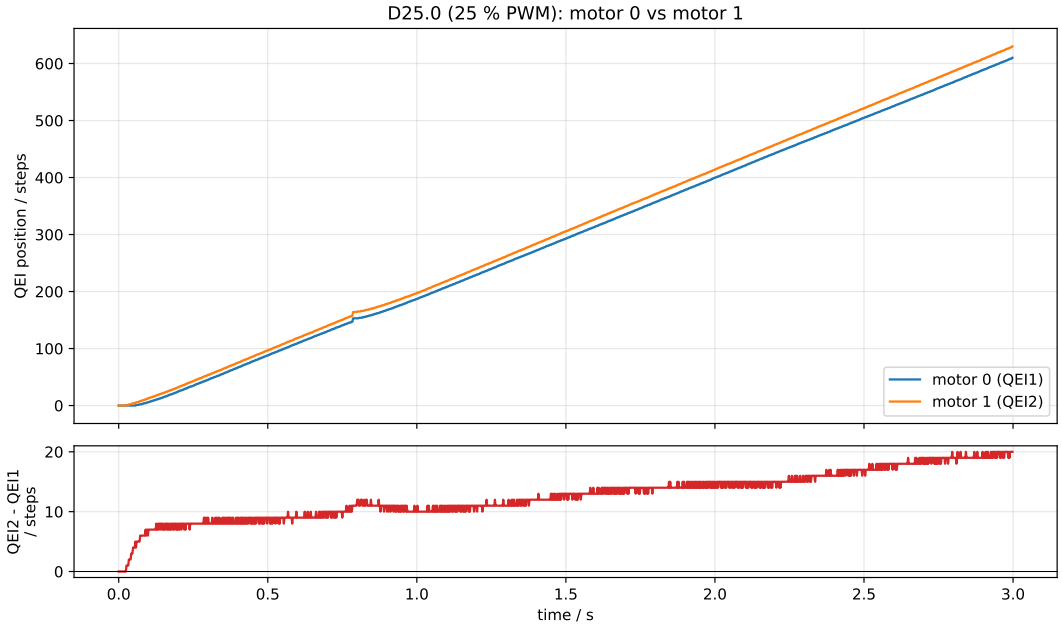


Abbildung 2.1.: Per-motor QEI step response at 25 % PWM. Top: motor 0 (`QEI1`) and motor 1 (`QEI2`) on a common axis; bottom: pointwise difference QEI2 − QEI1. The brief deviation around $t \approx 0{,}75\,\mathrm{s}$ corresponds to the single-sample QEI jump on motor 0 discussed in the text.

2.2. (f) Analysis of the RetBot's drive train

The goal of the analysis is to quantitatively describe the drive dynamics based on the recorded QEI signals (positions) and the velocities derived from them. The QEI time series were preprocessed for each duty cycle (uniform time base, offset correction), and the velocity was determined as a smoothed time derivative; this reduces high-frequency quantization and differentiation noise without significantly distorting the transient response. Because the QEI counter is integer-quantized, a naive forward-difference estimate of $\dot{\theta}$ is dominated by single-step quantization jumps. To suppress this without flattening the step response, a *Savitzky–Golay* filter with derivative order $d = 1$, polynomial order $p = 3$, and centered window length $N = 41$ samples ($\approx 82\,\text{ms}$ at $f_s = 500\,\text{Hz}$) is applied to each QEI time series (see Appendix C.2 for the general form, equation (C.6)). The same filter family with wider windows is used for the higher derivatives reported in the appendix; the velocity window is the narrowest because the velocity transient contains the highest-frequency content of physical interest in this experiment.
Figure 2.2 shows the QEI step response per drive train with the four duty cycles overlaid. Both wheels reach a near-constant slope (= angular velocity) within the first $\sim 500\,\text{ms}$ at 100 % PWM, and the steady-state slope scales close to linearly with the commanded duty cycle. Figure 2.3 reproduces the same data with motor 0 (dashed) and motor 1 (solid) overlaid per duty cycle and the difference QEI2 − QEI1 shaded; motor 1 (`QEI2`) consistently leads motor 0 by 2–3 % across all four conditions. Figure 2.4 shows the smoothed angular velocity for the same four conditions per drive train.

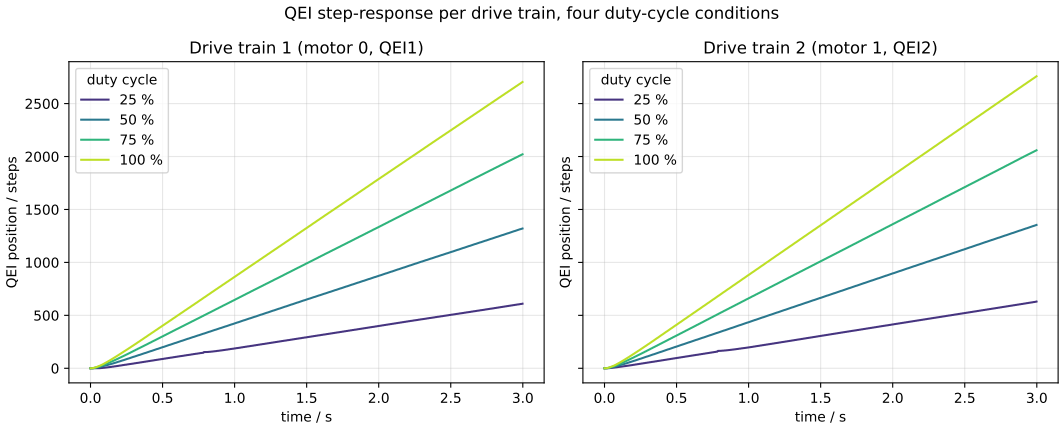


Abbildung 2.2.: QEI position step response per drive train, four duty-cycle conditions overlaid (25 %, 50 %, 75 %, 100 %). Left: motor 0 (`QEI1`); right: motor 1 (`QEI2`). Same vertical scale on both panels for direct comparison.

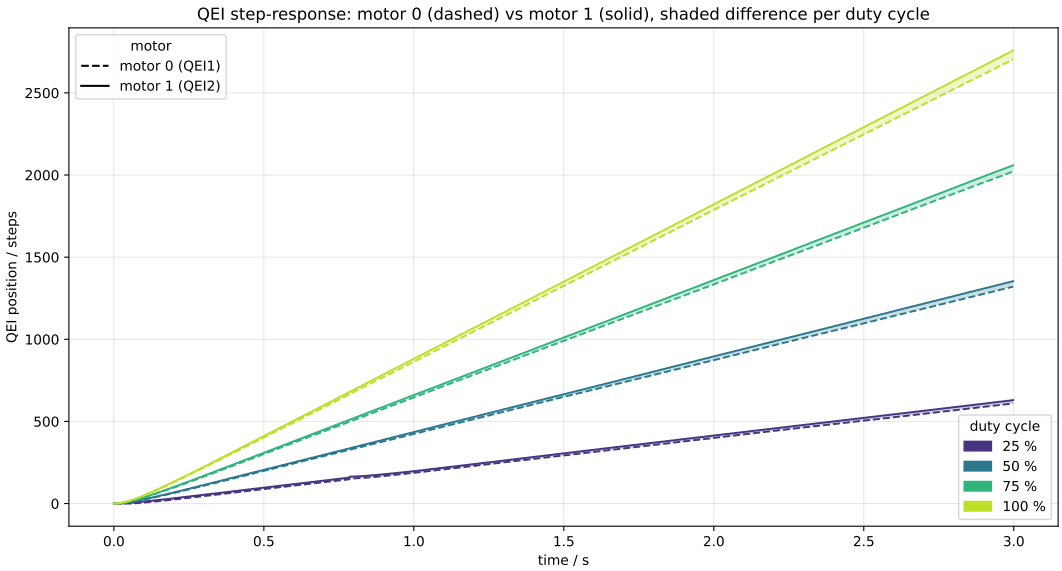


Abbildung 2.3.: QEI position step response, motor 0 (dashed) and motor 1 (solid) overlaid in the same panel for each duty cycle. The shaded band between each motor pair is the pointwise difference QEI2 − QEI1; the same color identifies the same duty cycle in lines and fill. Motor 1 leads motor 0 in every condition, by an amount that grows roughly linearly with travel.

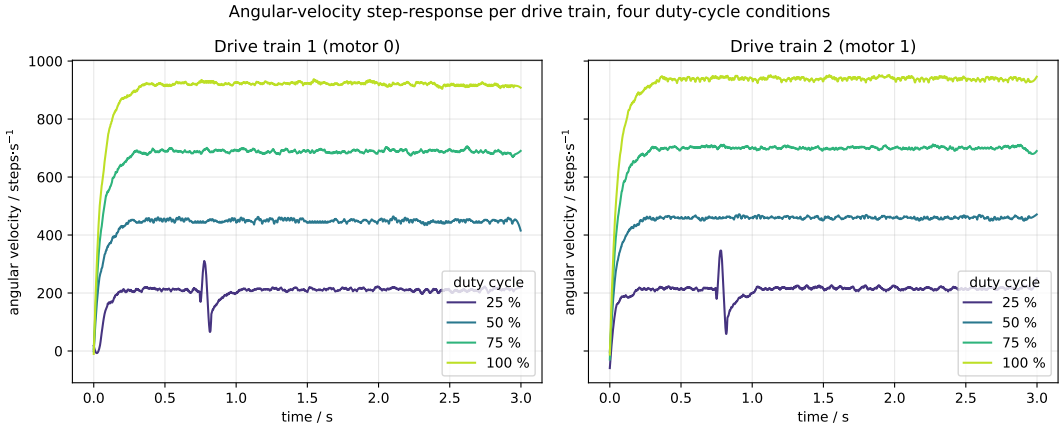


Abbildung 2.4.: Angular-velocity step response per drive train, computed as the Savitzky–Golay smoothed first derivative of the QEI position (equation (C.6) with d=1, p=3, N=41). Left: motor 0; right: motor 1; four duty-cycle conditions overlaid.

2.2.1. Conclusion

As part of the experiment, the system identification of the EV3 powertrain was successfully carried out, and the fundamental behavior of pulse width modulation (PWM) was experimentally validated. As the objectives and evaluation show, the direct relationship between duty cycle and motor behavior was demonstrated: the various duty cycles (25 %, 50 %, 75 %, 100 %) led to corresponding changes in output power via variations in the average motor voltage and resulted in clearly distinguishable, approximately proportional steady-state speeds. Furthermore, the position and velocity curves illustrated the real physical influences on the system. The observed asymmetry between the left and right motors, which led to a left drift, could be attributed to hardware tolerances. Despite occasional discontinuities in data acquisition (see QEI artifact during the 25 % measurement), the smoothed step responses provide a valid data basis.

A Appendix Frontmatter

A.1. Declarations

1. Wir erkläre hiermit, dass wir dieses Dokument selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet haben.

Regensburg, den 2026-05-01,

”	Marius Off	Matrikelnummer: 3416002,
”	Janis Graf	Matrikelnummer: 3377251

Ort, Datum und Unterschrift

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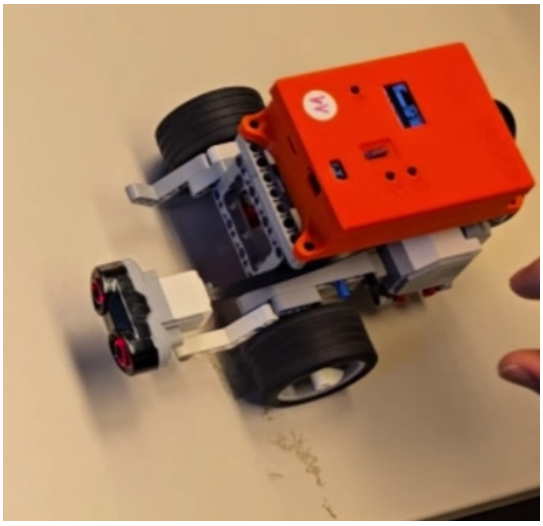
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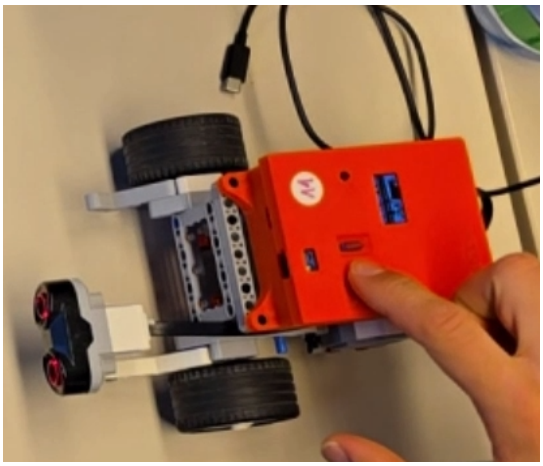
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B Appendix Code and Resources

B.1. Video resources



25% duty cycle
`https://youtu.be/q9igvjXbKiY`



75% duty cycle
`https://youtu.be/n6qsIMwFsFM`

Abbildung B.1.: Videos of the RetBot moving at two duty cycles.

B.2. Code used for the analysis of the Preperations

Editorial note — Source code omitted

The full source code for the analysis (a Jupyter notebook of this time approximately 1,500 lines) is *not* included in this appendix. Including it as a verbatim listing caused major build instability again and was therefore deliberately omitted.
The complete source can be obtained by either of the following means:

- By email request to `janis.graf@st.oth-regensburg.de`
- Via requesting access to the GitHub repository: `https://github.com/janisgra/RTP-T3-SoSe2026`

C Appendix extra calculations

Editorial note — out-of-scope appendix

This chapter exists because I cannot read instructions and somehow thought that it would be interesting to calculate the movement and rotation of the robot based on the QEI data. The contents are not part of task (f); they are kept here as supplementary calculations and may be safely skipped.

C.1. Differential-drive kinematics from QEI

Under the standard *differential-drive / no-slip* assumption the body-frame trajectory of the RetBot can be reconstructed from the two QEI counters alone. The convention used here is `QEI1` = motor 0 = left wheel, `QEI2` = motor 1 = right wheel; the wheel radius is r, the wheel separation L, and CPR the encoder counts per wheel revolution (after the gearbox).
None of these three mechanical parameters was measured directly for the specific RetBot used in this experiment, so the calculations below adopt nominal values for a typical EV3 educational chassis,

$$r = 28\,\text{mm}, \qquad L = 120\,\text{mm}, \qquad \text{CPR} = 360,$$

corresponding to a 56 mm nominal wheel diameter, a 12 cm center-to-center wheel separation, and the EV3 large-servo encoder's one-count-per-degree resolution at the output shaft. The absolute distances and angles reported in this section should therefore be read as order-of-magnitude estimates; the qualitative shape of the trajectories is unaffected by these parameter values. The encoder counts are first mapped to wheel angles and per-wheel arc lengths,

$$\theta_{L,R}(t) = \frac{\text{steps}_{L,R}(t)}{\text{CPR}}\,2\pi, \qquad s_{L,R}(t) = r\,\theta_{L,R}(t). \tag{C.1}$$

The body-frame forward path length and cumulative heading then follow as the average and length-normalized difference of the wheel arcs,

$$s(t) = \tfrac{1}{2}(s_R(t) + s_L(t))\,, \qquad \varphi(t) = \frac{s_R(t) - s_L(t)}{L}, \tag{C.2}$$

with $\varphi > 0$ counterclockwise viewed from above. The body-frame velocities are obtained as smoothed time derivatives, $v = \dot{s}$ and $\omega = \dot{\varphi}$, using the same Savitzky–Golay filter detailed in Appendix C.2. The resulting forward distance and heading change for the four main duty cycles are shown in Figure C.1.

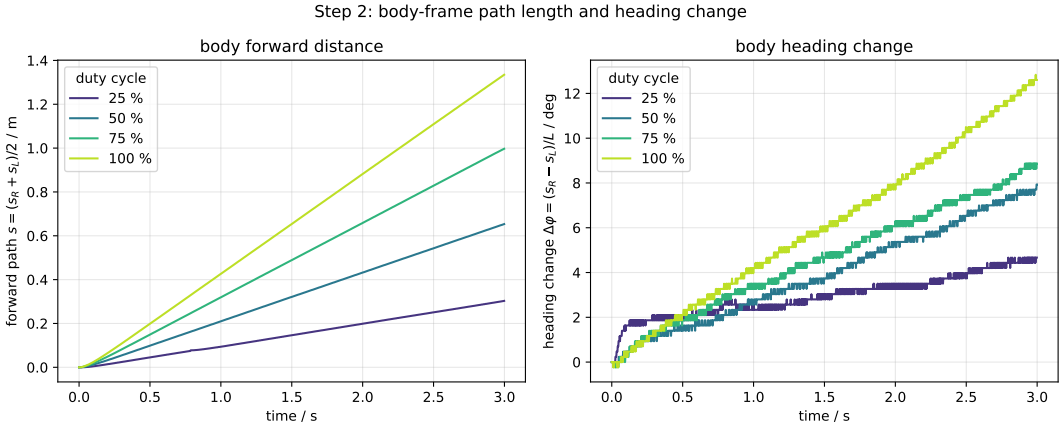


Abbildung C.1.: Body-frame forward path s (left) and cumulative heading change $\Delta\varphi$ (right) from (C.2) for the four main duty cycles. The non-zero heading is the integrated effect of the right wheel consistently leading the left.

The world-frame pose (x, y, φ) is obtained by discrete integration of the body-frame quantities. Treating each sample interval as a short circular arc of length Δs_k and turn angle $\Delta\varphi_k$, the midpoint-heading update is second-order accurate under non-zero curvature:

$$\varphi_k = \varphi_{k-1} + \Delta\varphi_k, \tag{C.3}$$

$$x_k = x_{k-1} + \Delta s_k \cos\Big(\varphi_{k-1} + \tfrac{1}{2}\Delta\varphi_k\Big), \tag{C.4}$$

$$y_k = y_{k-1} + \Delta s_k \sin\Big(\varphi_{k-1} + \tfrac{1}{2}\Delta\varphi_k\Big), \tag{C.5}$$

starting from $(x_0, y_0, \varphi_0) = (0, 0, 0)$. The reconstructed trajectories for the four constant-PWM drives plus the opposite-direction on-spot turn at $\pm 25\,\%$ are shown in Figure C.2.

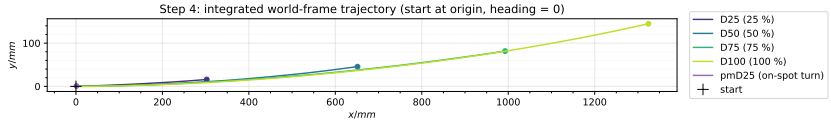


Abbildung C.2.: Reconstructed world-frame trajectories under (C.3)–(C.5) ($r = 28\,\text{mm}$, $L = 120\,\text{mm}$, CPR = 360). Straight drives at 25 % bis 100 % PWM curve slightly leftward (right wheel leads); the $\pm 25\,\%$ on-spot turn produces $\approx 282°$ of net rotation with $s \approx 0$.

The on-spot turn provides a clean self-consistency check for the kinematics: with wheel commands of opposite sign and equal magnitude the arcs satisfy $s_L \approx -s_R$, so by (C.2) the body-frame forward path collapses to $s \approx 0$ while φ accumulates linearly — exactly as observed (final $s = -0.1\,\text{mm}$, $\Delta\varphi = +281.9°$ over the 3 s log). Note that this consistency

check holds independently of the assumed r, L, and CPR: an absolute calibration of r against a measured reference distance would simply rescale s, and heading accuracy in turns is governed only by the ratio r/L.

C.2. Calculation of the derivatives and integrals of the motor position

The recorded QEI position $\theta(t)$ is integer-quantized at the encoder resolution; naive finite-difference differentiation therefore amplifies the sample-to-sample quantization noise by a factor that grows with derivative order. To obtain usable estimates of the higher derivatives a *Savitzky–Golay* filter is used: in a centered sliding window of N samples a polynomial of degree p is fitted in the least-squares sense, and the analytic derivative of order d of that polynomial is evaluated at the central sample,

$$\left.\frac{d^d\theta}{dt^d}\right|_{t_k} \;\approx\; \frac{1}{(\Delta t)^d} \sum_{i=-(N-1)/2}^{(N-1)/2} c_i^{(d,p,N)}\, \theta_{k+i}, \tag{C.6}$$

with $\Delta t = 1/f_s$ the sample spacing and $c_i^{(d,p,N)}$ the tabulated convolution coefficients. The constraint $N > p \geq d$ must hold; N is widened with d to keep the output smooth without distorting the step response:

quantity	order d	polyorder p	window N	window length
velocity	1	3	41	82 ms
acceleration	2	4	51	102 ms
jerk	3	5	71	142 ms

The motor *absement* $A(t) = \int_0^t \theta\, d\tau$ is computed by trapezoidal cumulative integration,

$$A(t_k) \;\approx\; \sum_{i=1}^{k} \tfrac{1}{2}\big(\theta_{i-1} + \theta_i\big)\, \Delta t, \tag{C.7}$$

which is $\mathcal{O}(\Delta t^2)$ accurate and does not amplify quantization noise. Figure C.3 shows the resulting chain for the 50 % PWM step response: each panel makes one stage of the absement → position → velocity → acceleration → jerk transformation visible on a common time axis, with both motors overlaid per panel.

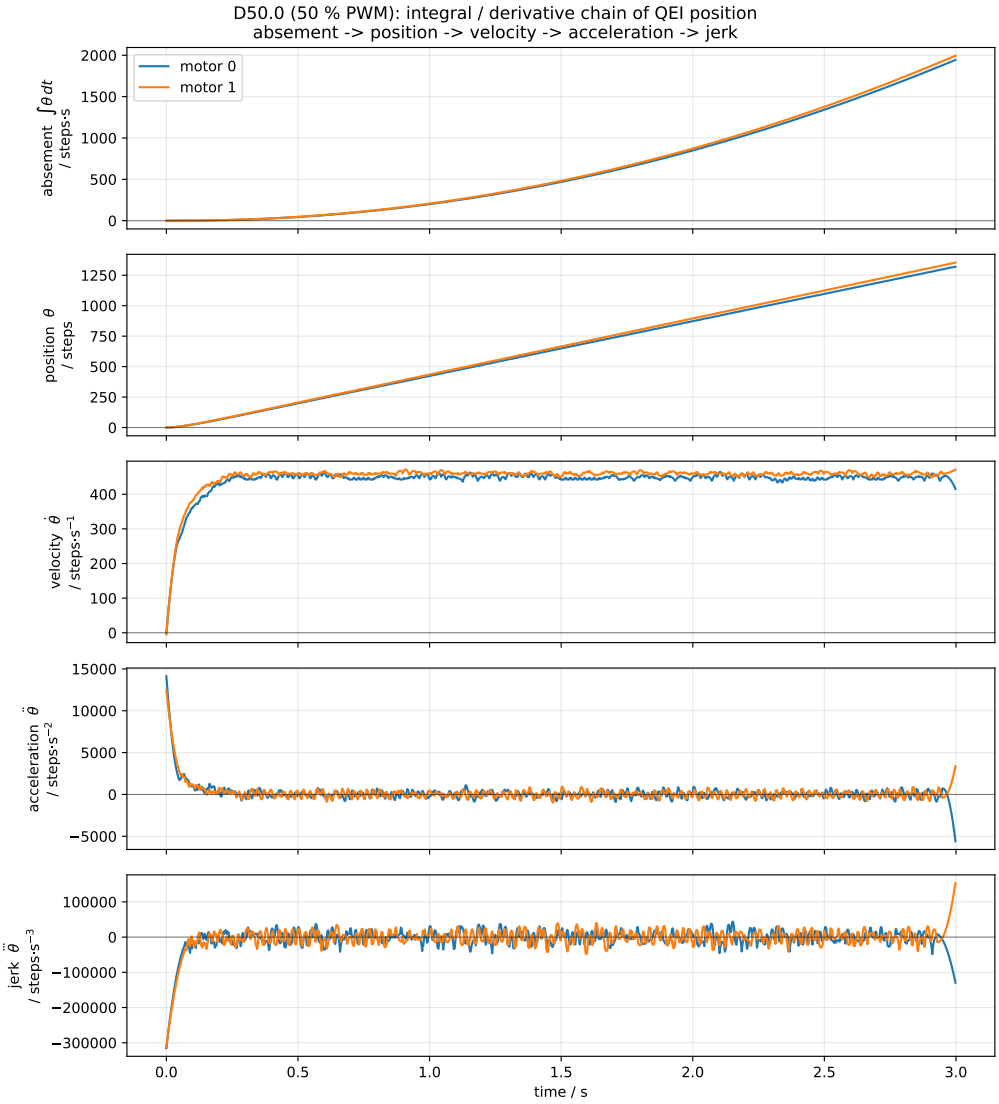


Abbildung C.3.: Integral / derivative chain of the QEI position for the 50 % PWM step response: absement, position, velocity, acceleration and jerk on a common time axis. Both motors overlaid per panel.

Literatur

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[2] *RS-232*. In: *Wikipedia*. 10. März 2026. URL: https://en.wikipedia.org/w/index.php?title=RS-232&oldid=1342757218 (besucht am 29. 04. 2026).

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